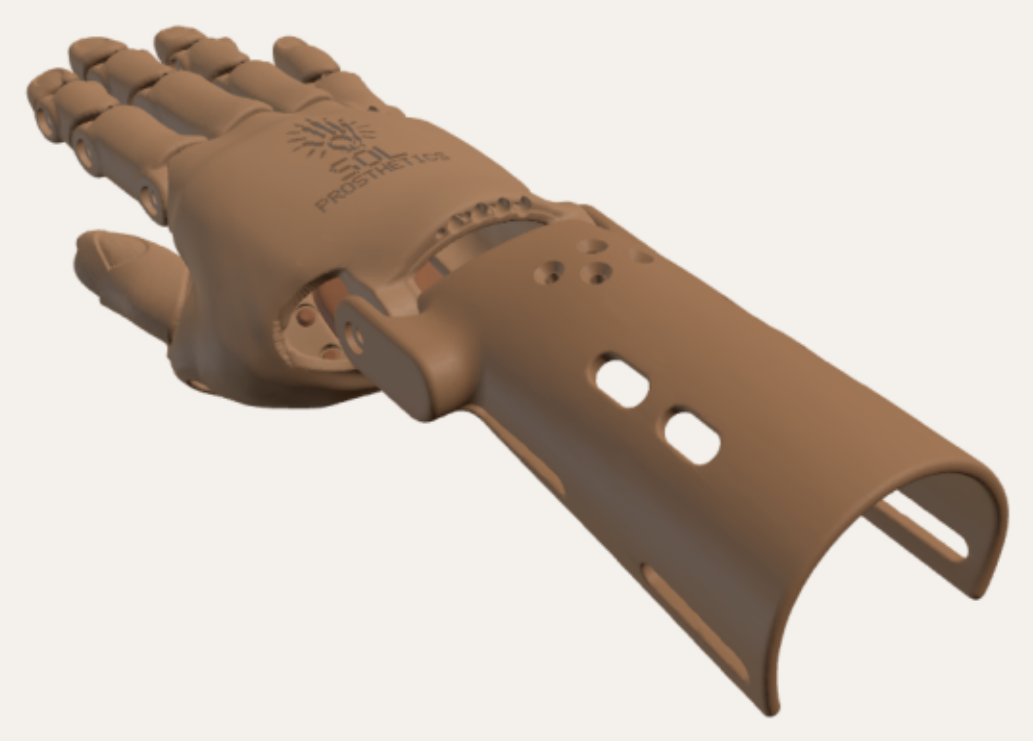
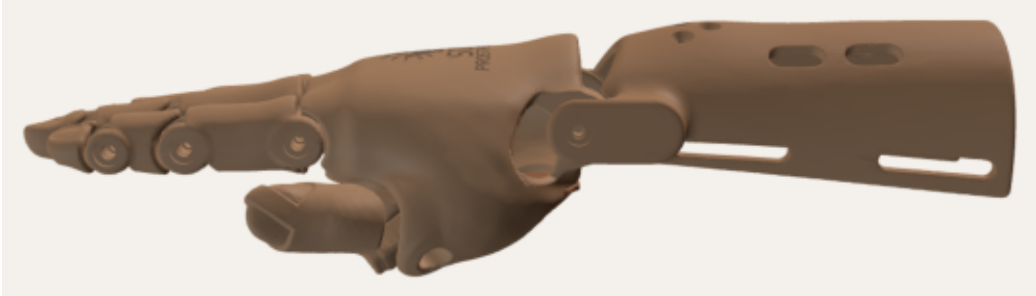
Bukang-liwayway v3.1g

Illustrated assembly and stringing guide

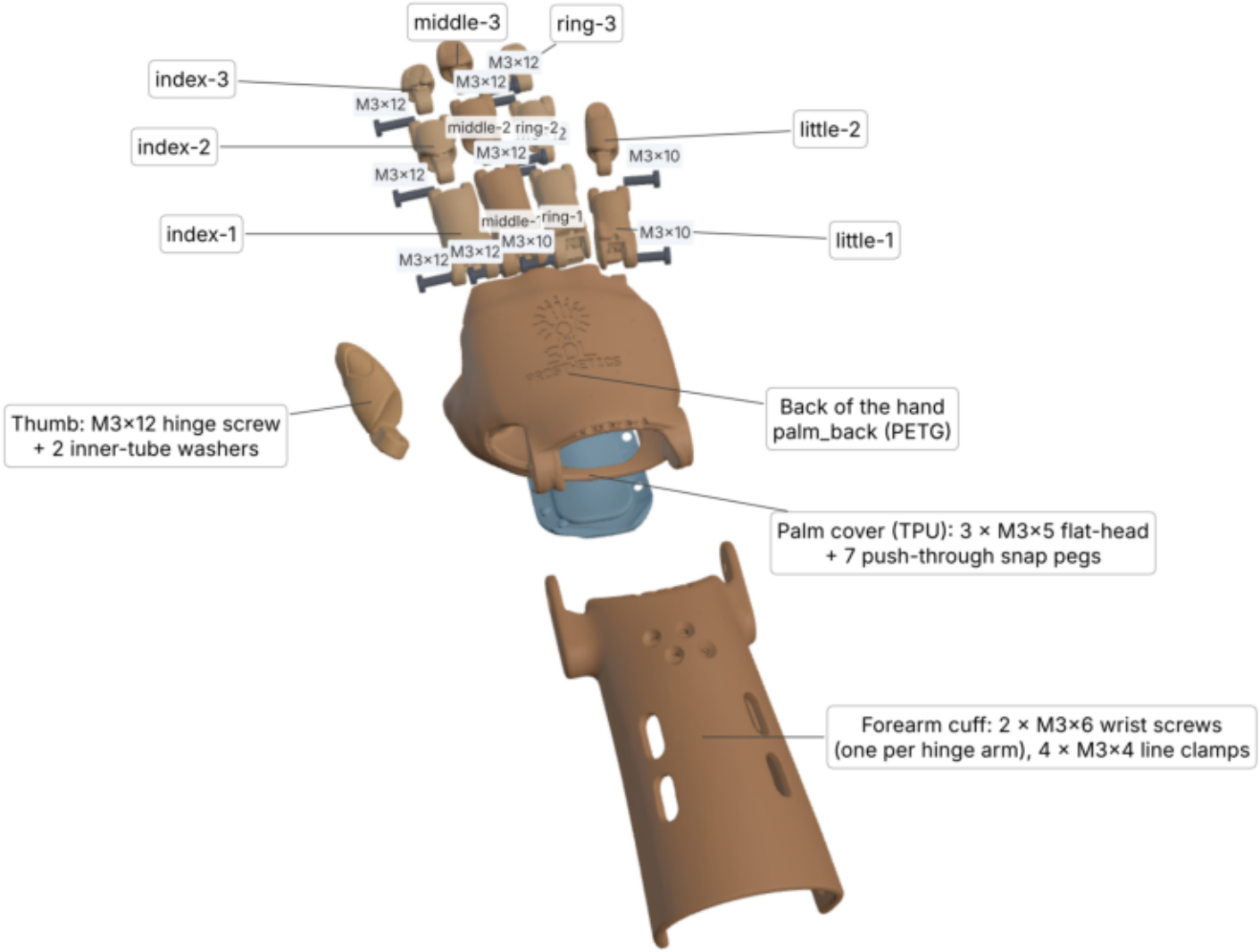
Use with the written Printing & Assembly Guide (parts list, print settings, tuning and care).





Order: 1 fingers · 2 fingers to palm · 3 thumb · 4 return cords · 5 flexor lines · 6 wrist · 7 tensioner · 8 grip pads · 9 palm cover · 10 straps and liner

Parts and screws — exploded view



1 Build each finger

Join segment 1 to 2, then 2 to 3. Each joint is a barrel in a cup held by one button-head screw. Put it in from the side with the round recess so the head sits flush. It threads into the plastic, no nut. Sizes are printed next to each screw in the picture.

2 Fingers onto the palm

Order: middle, index, ring, little, so every screw stays reachable. Slide segment 1's cup over its knuckle tongue and screw in from the recessed side.

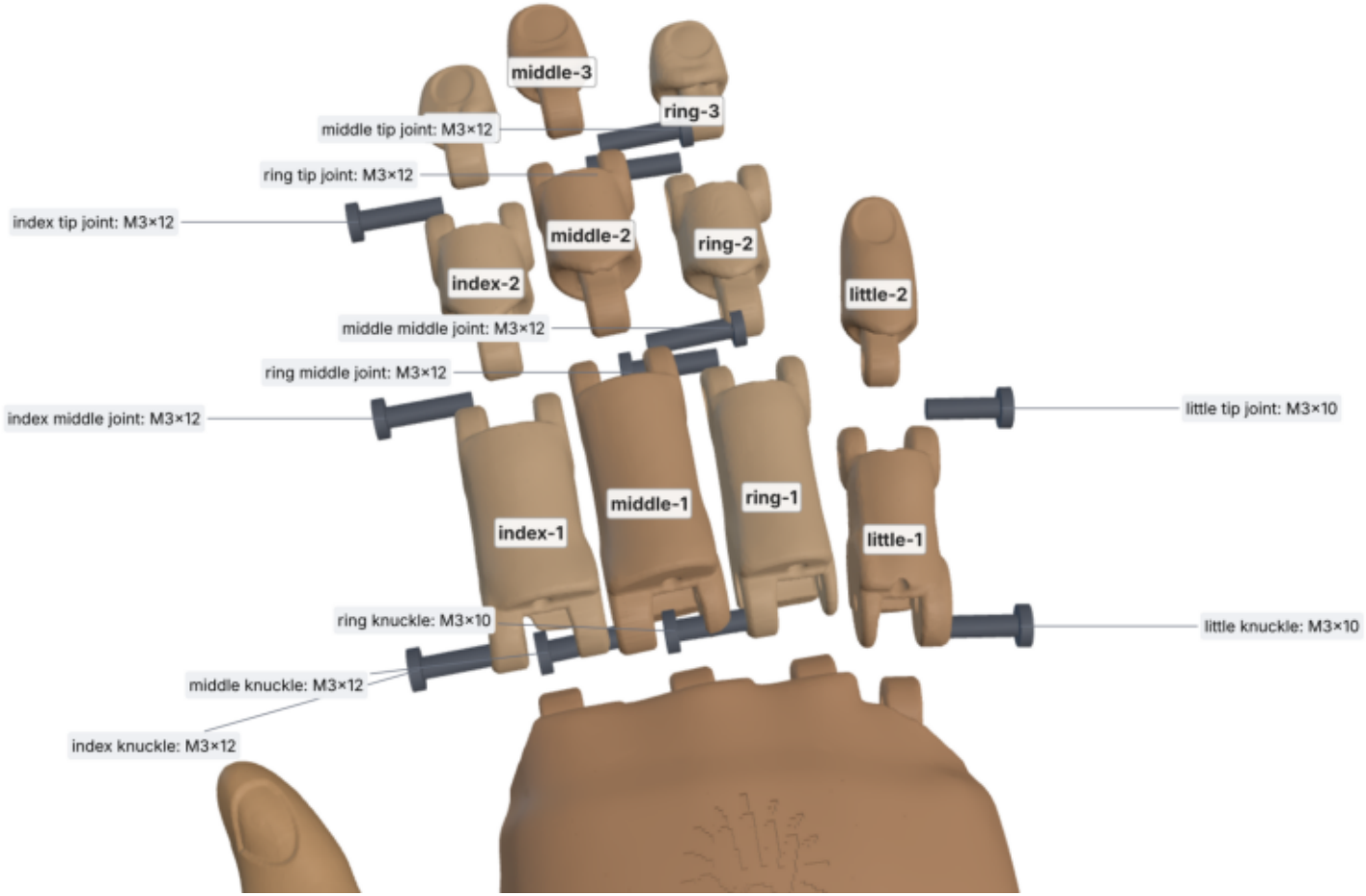
3 Thumb

An inner-tube washer on each side of the thumb tongue, into the slotted barrel, M3×12. Swing it to the grip and tighten until it holds.

Tighten

Only until the joint swings freely with no side wobble.

Fingers — which screw goes where



Screws per joint

Every joint uses one M3 button-head screw. Lengths: M3×12 for all index, middle and ring joints except the ring knuckle (M3×10); M3×10 for both little-finger joints. Thumb hinge: M3×12.

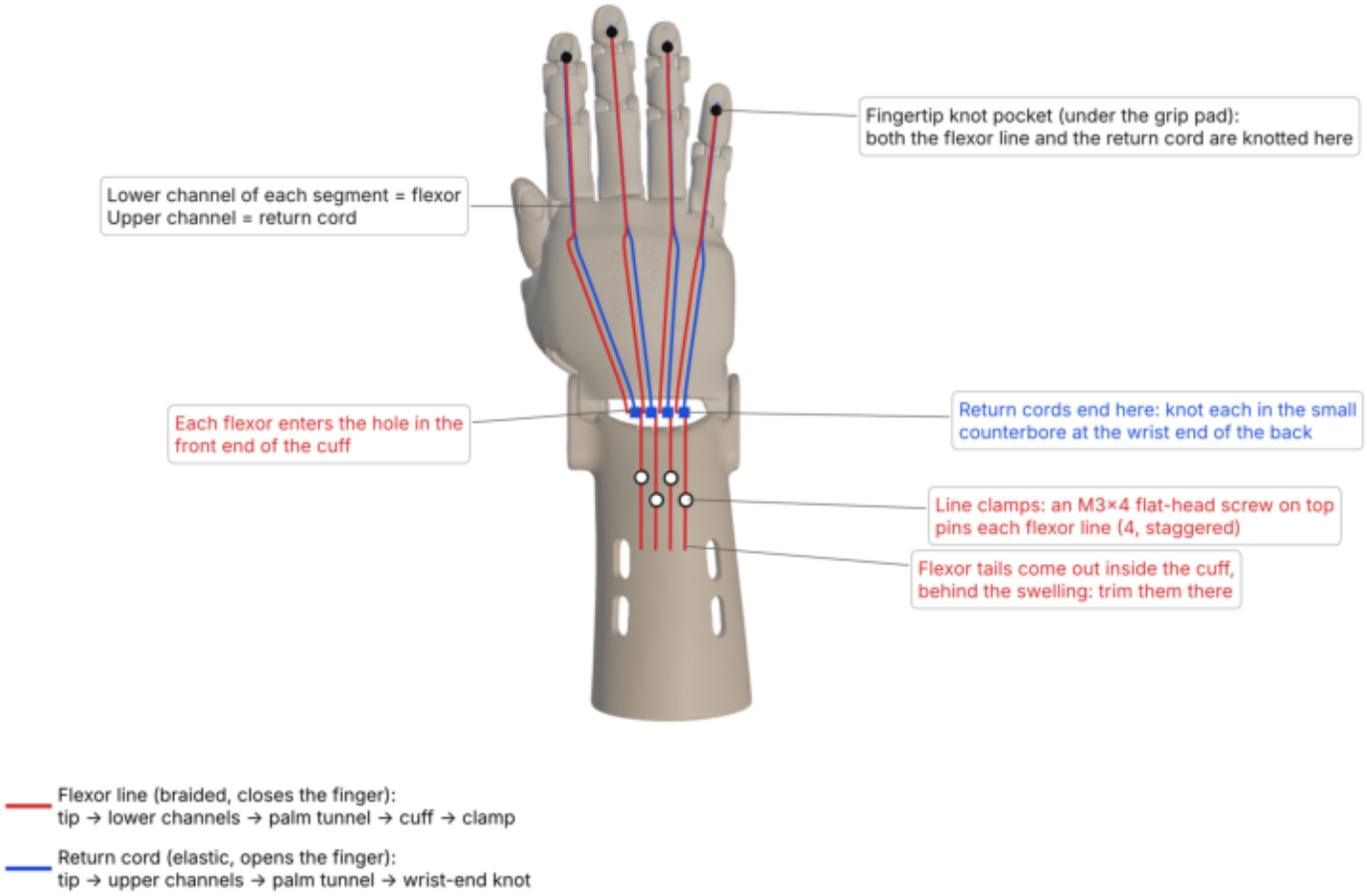
Head side

Each screw goes in from the side with the round recess, as drawn. The far side is a 2.6 mm pilot hole that goes all the way through; the screw cuts its own thread.

Check

Before stringing, bend every joint by hand: it should swing freely and stop just past straight.

Stringing — the four fingers



4 Return cords (blue)

Knot the 1 mm elastic in the fingertip knot pocket. Thread it through the UPPER channel of each segment, then the tunnel in the back of the hand. Knot it in the small counterbore at the wrist end, just snug enough to hold the finger open.

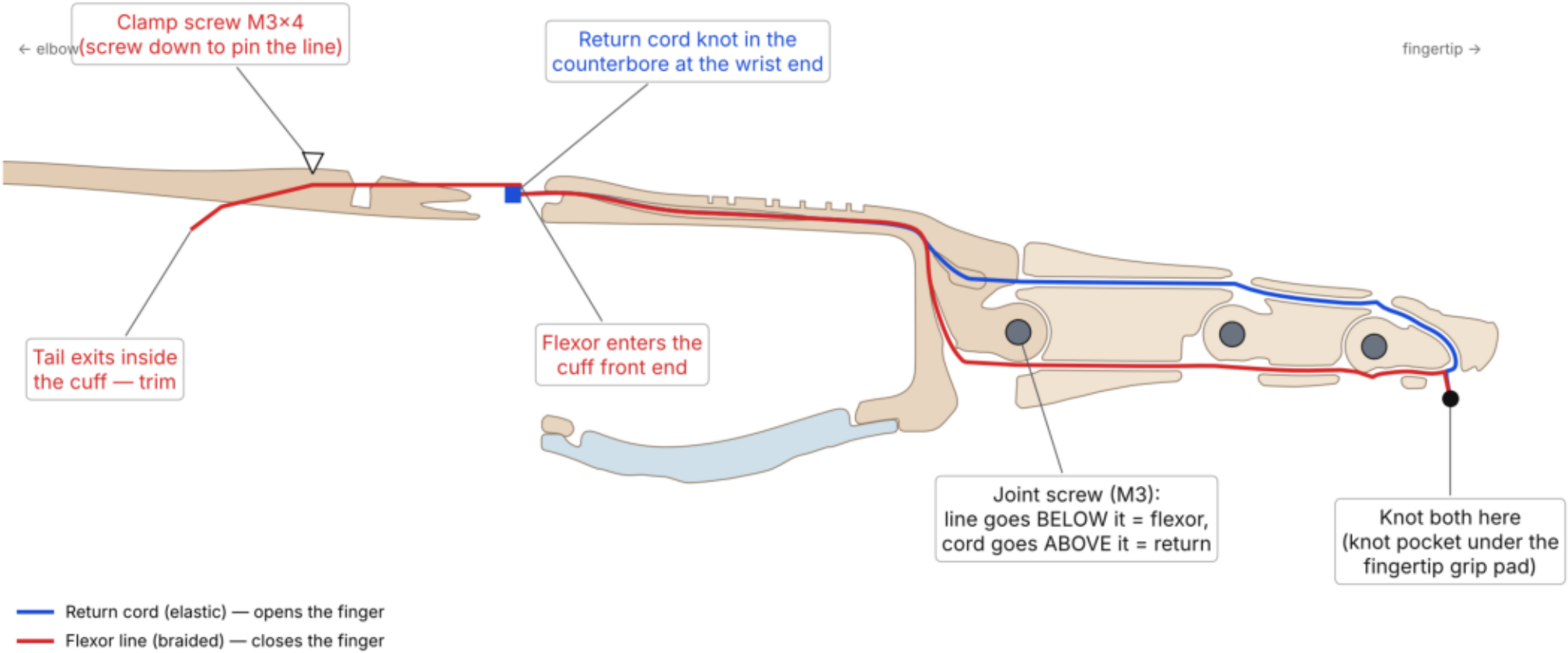
5 Flexor lines (red)

Knot the braided line in the same fingertip pocket and seal the knot. Thread it through the LOWER channel of each segment, into the knuckle block and through its tunnel to the wrist end. A thin wire or long needle helps.

Tip

Do one finger completely before the next, and label the line ends with tape so you know which line belongs to which finger.

Stringing — side view of one finger (middle finger, cut lengthwise; the others are the same)



Which channel is which

Flexor = below the joint screws (palm side). Return = above them (back of the hand). Pulling the flexor bends every joint; the elastic springs them straight.

6 Wrist

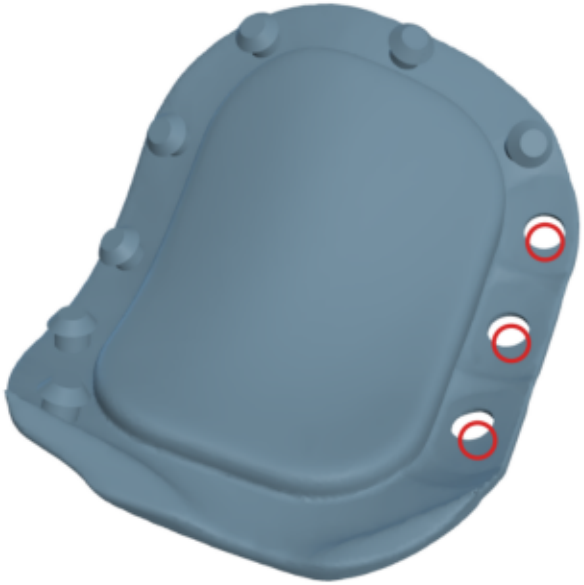
Join the back of the hand to the cuff with two M3×6 screws through the cuff's hinge arms.

7 Tensioner

Feed each flexor into its hole in the front end of the cuff. With the wrist straight and fingers open, pull snug and screw down its M3×4 clamp until flush. Trim the tail inside the cuff. Bend the wrist down; adjust until all four fingers close together.

Grip pads, palm cover, straps

Palm cover (TPU), inside face



Red rings: the 3 hinge screws (M3×5 flat-head), little-finger edge. The 7 mushroom pegs under the lip push through the 7 holes in the step.

Back of the hand, palm side: the step



Lay the cover in this step, screw the hinge edge, then press each peg through its hole until the head pops out on the inside (click).

8 Grip pads

Inner-tube pads into the 5 fingertip recesses with contact cement: thin film on both, wait about 10 minutes, press firmly.

9 Palm cover

Lay the hinge edge in the step and fit the three M3×5 flat-head screws until flush. Don't overtighten. Fold the cover shut and press over each of the 7 pegs until it pops through (click). To open, peel the edge back firmly.

10 Straps and liner

Sew the two straps on through the cuff slots, fit the liner, sand any edge that touches skin.